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Machine learning analysis of survival outcomes in breast cancer patients treated with chemotherapy, hormone therapy, surgery, and radiotherapy

Eyachew Misganew Tegaw[1]✉ & Betelhem Bizuneh Asfaw[2]

Breast cancer continues to be a leading cause of death among women in the world. The prediction of survival outcomes based on treatment modalities, i.e., chemotherapy, hormone therapy, surgery, and radiation therapy is an essential step towards personalization in treatment planning. However, Machine Learning (ML) models may improve these predictions by investigating intricate relationships between clinical variables and survival. This study investigates the performance of several ML models to predict survival rate in patients undergoing diverse breast cancer treatments i.e., chemotherapy, hormone therapy, surgery and radiation using multiple clinical parameters. The dataset consisted of 5000 samples and turned into downloaded from Kaggle. The models assessed blanketed Support Vector Machines (SVM), K-Nearest Neighbor (KNN), AdaBoost, Gradient Boosting, Random Forest, Gaussian Naive Bayes, Logistic Regression, Extreme Gradient Boosting (XG boost), and Decision tree. Performance of the models was assessed using parameters such as Accuracy, Precision, Recall, F1-Score and Area under the Receiver Operating Characteristic Curve (AUC-ROC). SHAP (SHapley Additive exPlanations) values analysis was done to provide an explanation for the impact of a feature on model predictions using Waterfall and Beeswarm plots. Anticipated baseline (E(f(x))) were in comparison to the predictions (f(x)) for each therapy group. Performance of Gradient Boosting was shown to be the best with an Accuracy: 0.972, Precision: 0.973, Recall: 0.972, F1-Score: 0.973, and AUC-ROC Score: 0.997. Chemotherapy had a notably bad impact on survival, with an f(x) of -0.274 and an E(f(x)) of -0.025. Hormone therapy showed the maximum detrimental effect on survival, with an f(x) of -0.408. Surgical operation had an extraordinarily impartial impact (f(x) = -0.041), even as radiation therapy positively impacted survival results with an f(x) of 0.22. Gradient Boosting was the most predictive algorithm for breast cancer survival outcomes. This SHAP-primarily based analysis provides a complete knowledge of ways one-of-a-kind treatments have an effect on survival predictions in breast cancer patients. Radiation therapy indicates the most tremendous effect on survival, whilst hormone therapy reveals the maximum poor effect. Future studies need to explore personalized treatment strategies that comprise these insights to enhance patient effects.

Keywords Machine learning, Breast Cancer, Survival prediction, SHAP analysis, Hormone therapy, Chemotherapy, Surgery, Radiation therapy

Abbreviations

AUC-ROC	Area under the receiver operating characteristic curve
BMI	Body mass index
ML	Machine learning
KNN	K-Nearest neighbor
SHAP	SHapley additive exPlanations
SMOTE	Synthetic minority oversampling technique

[1]Department of Physics, College of Natural and Computational Sciences, Debre Tabor University, 272, Debre Tabor, Ethiopia. [2]Department of Health System Management and Health Economics, School of Public Health, College of Medicine and Health Sciences, Bahir Dar University, Bahir Dar, Ethiopia. ✉email: eyachew2003@gmail.com

SVM	Support vector machines
XG boost	Extreme gradient boosting

Breast cancer is a major common form of cancer among females globally and accounts for a vast number of overall morbidity and mortality related to the disease[1]. Chemotherapy, hormone therapy, surgery, and radiation therapy are just some of the treatment options available for breast cancer. However, their relative efficacy may differ depending on individual patient characteristics, including biological tumor type or genetic mutations[2]. The accurate prediction of survival outcomes for patients treated with this treatment would help to optimize therapeutic decisions and clinical prognosis in these patient populations.

Conventional prognostic models in breast cancer using linear statistical methods might not be able to identify complex interactions between multiple clinical factors simultaneously[3]. New opportunities are offered by Machine learning (ML) in oncology that allows for the evaluation of large datasets to discover patterns and predictive models. These models have demonstrated the ability to predict outcomes relying on patient and tumor covariates that traditional methods can fail to identify, improving precision medicine[4,5].

In this study, we developed a ML model to predict breast cancer patients' survival after getting chemotherapy, hormone therapy, surgery, and radiation therapy. SHAP (SHapley Additive exPlanations) values, a tool from the field of explainable artificial intelligence, offer an intuitive method to interpret complex ML models. These values measure the contribution of the features to a model's prediction for a character patient, thereby delivering personalized insights into therapy efficacy. We used SHAP values to examine the impact of different breast cancer treatment options on survival consequences, incorporating key clinical variables including age, tumor size, tumor type, lymph node status, hormone receptor status, genetic mutation, follow-up duration, blood pressure, and Body Mass Index (BMI) from a dataset containing 5000 samples to be had at Kaggle (https://www.kaggle.com/datasets/smmmmmmmmmmmm/breast-cancer-analysis). This study differentiates itself by using SHAP values to interpret how the clinical features and different treatment modalities impact breast cancer survival predictions independently. In contrast to existing studies limited on model performance only, we link ML with SHAP analysis technique for transparent interpretation of single-level survival benefits due to chemotherapy, hormone therapy and surgery as well radiation therapy.

Methods

In this study, each patient was treated with a single modality of treatment (chemotherapy, hormone therapy, surgery, or radiation therapy), as per the dataset structure. This allowed for an undiluted assessment of the independent effect of each treatment on survival outcomes. We acknowledge, however, that in clinical practice, the majority of breast cancer patients are treated with combination therapies. Therefore, our findings should be interpreted with the caveat that they reflect single-modality treatment effects.

Data source and preprocessing

The dataset for this study was obtained from the Kaggle platform (https://www.kaggle.com/datasets/smmmmmmmmmmmm/breast-cancer-analysis) and contains clinical data on 5,000 patients with breast cancer. We hereby declare that the Kaggle dataset for this work is open-access and fully anonymized. This, in turn, means that patients' privacy will not be breached since there is no personal identifiable information in the dataset. We also affirm that the use of data does not violate any pertinent ethical standards as it is a secondary form of data collected anonymously. Its components include treatments (chemotherapy, hormone therapy, surgery, and radiation therapy), follow-up duration, blood pressure, BMI, genetic mutations, lymph node status, tumor size, tumor type, patient age, and hormone receptor status. The target variable is the state of survival, whether it is alive or deceased. No values were missing from the dataset. The data was preprocessed using a number of methods. First, the numerical features (age, tumor size, follow-up duration, blood pressure, and BMI) were normalized using Min-Max scaling to ensure that they were within a common range. The reason for this is that ML techniques such as Support Vector Machines (SVM) and Gradient Boosting work better with scaled inputs. The second step was to use label encoding to translate categorical features (tumor type, lymph node status, hormone receptor status, genetic mutation, treatment, and survival status) into a numerical representation. A numeric value was allocated to each category. It is an important step because the majority of ML models use numerical inputs to do calculations. The Interquartile Range (IQR) approach was used in step three to find outliers in numerical data. Values that were deemed outliers that fell more than 1.5 times the IQR were either eliminated or modified to avoid distorting the findings[6,7]. The survival status was skewed towards the 'Alive' class, proving that the dataset was class imbalanced. This was addressed in step four by using the Synthetic Minority Oversampling Technique (SMOTE), which balanced the training dataset by creating synthetic samples for the minority class ('Deceased').

Descriptive statistics

Descriptive statistics were calculated for all clinical and demographic variables in the 5,000 breast cancer patient dataset. For numerical variables such as age, tumor size, follow-up time, blood pressure, and BMI, means, standard deviations, and ranges (minimum and maximum values) were computed. For categorical variables such as tumor type, lymph node status, hormone receptor status, genetic mutation status, survival status, and treatment type, frequencies and percentages were utilized to describe them. Frequencies of the types of treatment (chemotherapy, hormone therapy, surgery, and radiation) and survival status (dead or alive) were also analyzed to get a general view of the patient population. These statistics were computed in order to get a clear view of the nature of the dataset before further analysis.

Model development

Using the pre-processed dataset, we trained and optimized nine ML models: AdaBoost, Gaussian Naive Bayes, Random Forest, SVM, KNN, Logistic Regression, Extreme Gradient Boosting (XGBoost), and Decision Tree[8]. The parameter optimization for each of the models is shown in Table 1.

To resolve the class imbalance in the training data, we used SMOTE after splitting the dataset into an 80% training set and a 20% test set[9].

10-fold Cross-Validation was employed to prevent overfitting and Hyperparameter tuning with Grid Search was used to Control Complexity.

Performance metrics

The performance of the models was evaluated using metrics for performance assessment such as Accuracy, Precision, Recall, F Score, and AUC-ROC. Accuracy measures the proportion of correct predictions made by the model. Precision indicates how many true positive results are among all predicted results. Recall shows how well the model identifies cases. The F1-Score provides a measure between Precision and Recall by taking their harmonic mean into account. The AUC-ROC curve assesses how well the model can differentiate between negative categories at cutoff points; higher scores indicate better performance[10]. This result was used to generate a heatmap that visually contrasts the models on these criteria. Since the Gradient Boosting model was best performing, it has been further evaluated using SHAP on how to interpret feature contributions when predicting survival.

SHAP analysis

We used SHAP (SHapley Additive exPlanations) values, which measure each feature's contribution to the predicted survival result (f(x)), to analyze the predictions of the top-performing model further. The impact of individual variables on survival forecasts for particular patients was visualized using the SHAP Waterfall (which describes the contribution of each feature to a single prediction) and Beeswarm (illustrates the spread of SHAP values for all features over an entire dataset) plots[11]. The predicted result, denoted as f(x), represents the model's estimation of a patient's survival probability based on their clinical characteristics. This was compared to the baseline prediction E(f(x)), which is the average survival probability across all patients. The difference between f(x) and E(f(x)) illustrates how each feature and treatment modalities (chemotherapy, hormone therapy, surgery, and radiation therapy) shifts a patient's predicted survival outcome compared to the baseline.

Results

Descriptive statistics

Descriptive statistics were calculated for all clinical and demographic variables in the 5,000 breast cancer patient dataset as shown in Table 2 and Figs. 1, 2, 3 and 4.

The distribution of overall survival status, which is a good summary of the general survival experience among the study population is presented in Fig. 1. The graph indicates that 51.92% of the patients survived, while 48.08% had died. This nearly even ratio provides an initial insight into the survival rate in the study, suggesting that the population is in close to equal ratio between survivors and non-survivors.

The percentage distribution of the different treatments that the patients received is presented as shown in Fig. 2. The treatments include hormone therapy, surgery, radiation, and chemotherapy. The most common treatment is hormone therapy with 25.96%, closely followed by surgery (25.50%), radiation (24.46%), and chemotherapy (24.08%). The relatively equal distribution of the treatments means that there is no treatment that dominates the others, providing a balanced view of the treatments in the study.

ML models	Parameters and Description
Gradient Boosting	loss='log_loss', learning_rate = 0.1, n_estimators = 100, subsample = 1.0, criterion='friedman_mse', min_samples_split = 2, min_samples_leaf = 1, max_depth = 3, random_state = None, max_features = None, verbose = 0, max_leaf_nodes = None
Random Forest	n_estimators = 100, criterion='gini', max_depth = None, min_samples_split = 2, min_samples_leaf = 1, max_features='sqrt', max_leaf_nodes = None, min_impurity_decrease = 0.0, random_state = None, verbose = 0, max_samples = None
KNN	n_neighbors = 5, metric_params = None
AdaBoost	estimator = None, n_estimators = 50, learning_rate = 1.0, random_state = None
SVM	C = 1.0, kernel='rbf', probability = False, class_weight = None, verbose = False, max_iter=-1, random_state = None
Logistic Regression	penalty='l2', class_weight = None, random_state = None, max_iter = 100, multi_class='auto', verbose = 0
Decision Tree	max_depth = None, min_samples_split = 2, min_samples_leaf = 1, max_features = None, random_state = None, max_leaf_nodes = None, class_weight = None
XGBoost	booster='gbtree', verbosity = 1, validate_parameters = 1, use_label_encoder = False, objective='binary: logistic', n_estimators = 100, max_depth = 6, learning_rate = 0.3, random_state = None
Gaussian NB	var_smoothing = 1e-9

Table 1. ML model parameter configurations. This table demonstrates the parameter values for the various ML models used in the study. The table provides details for Gradient Boosting, Random Forest, KNN, AdaBoost, SVM, Logistic Regression, Decision Tree, XGBoost, and Gaussian Naïve Bayes. Each model's parameters such as the number of estimators, learning rate, depth limits, and feature selection limit are provided to maintain transparency within the modeling.

Variable	Count	Mean	Std Dev	Min	25%	50%	75%	Max
Age	5000	49.70	17.27	20.0	35.0	50.0	65.0	79.0
Tumor Size	5000	5.19	2.73	0.50	2.83	5.14	7.55	9.99
Tumor Type	5000	0.50	0.50	0.0	0.0	1.0	1.0	1.0
Lymph Node Status	5000	0.50	0.50	0.0	0.0	0.0	1.0	1.0
Hormone Receptor Status	5000	0.99	0.81	0.0	0.0	1.0	2.0	2.0
Genetic Mutation	5000	0.99	0.81	0.0	0.0	1.0	2.0	2.0
Follow-Up Duration (Months)	5000	66.15	31.18	12.0	39.0	66.0	94.0	119.0
Blood Pressure (mmHg)	5000	129.42	28.64	80.0	105.0	130.0	154.0	179.0
BMI (kg/m^2)	5000	28.99	6.34	18.0	23.38	29.05	34.35	39.99
Survival Status (0 = Deceased, 1 = Alive)	5000	0.52	0.50	0.0	0.0	1.0	1.0	1.0

Table 2. Descriptive statistics of key variables. This table displays the descriptive statistics of important variables in the data set, including age, tumor size, tumor type, lymph node status, hormone receptor status, genetic mutation, Follow-Up duration, blood pressure, BMI, and survival status. The values given are count, mean, standard deviation, minimum, quartiles (25%, 50%, 75%), and maximum.

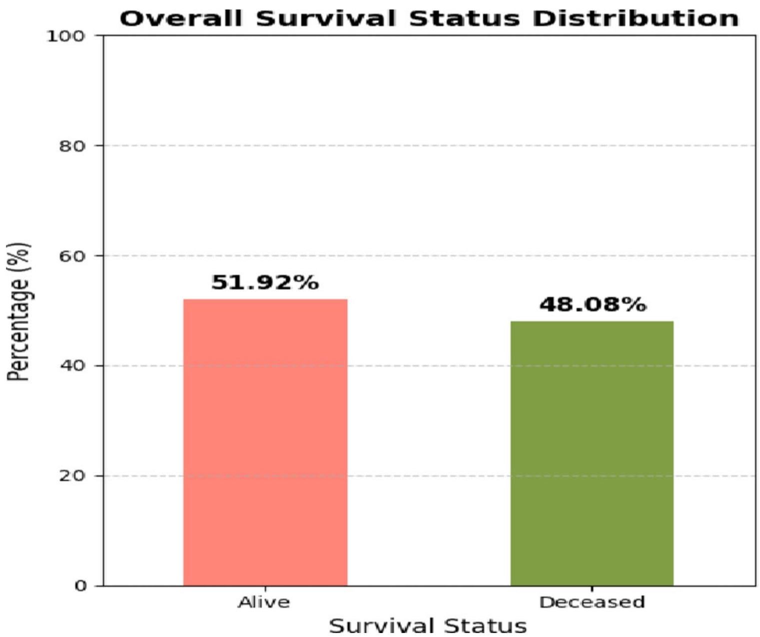

Fig. 1. Overall Survival Status Distribution. This graph presents the distribution of survival status, with 51.92% alive and 48.08% dead. It provides a good summary of the overall survival outcome in the studied population.

Survival status is splits into tumor types, benign versus malignant tumors as shown in Fig. 3. Among benign tumors, 25.96% are alive, and 23.66% are deceased. Among malignant tumors, 25.96% are alive and 24.42% have died. How closely survival rate approaches equality in the two forms of tumors illustrates that tumor type could not play a significant factor in survival status within this research and therefore exploration into other probable factors should be considered.

The survival outcomes by the type of treatment given to the patients is presented in Fig. 4. It compares the survival percentages for chemotherapy, hormone therapy, radiation, and surgery. Chemotherapy and hormone therapy have slightly better survival percentages, with 12.86% and 13.24% of the patients surviving, respectively. Radiation and surgery share similar survival and deceased rates, and this indicates that these treatments are equally effective from the viewpoint of survival outcomes. This analysis provides valuable information regarding relative effectiveness among different treatments, which can inform future treatment plans and decisions.

The statistical significance of the variables used as inputs of this study are calculated as shown in Table 3.

Class distribution

The target class exhibited an imbalance, consisting of 2077 cases of Alive and 1923 cases of Deceased before the application of SMOTE. Following the use of SMOTE, both classes achieved equilibrium with 2077 cases. This modification addresses the problem of class imbalance, which could result in suboptimal performance from both ML models if left uncorrected. The class distribution before and after SMOTE are shown in Fig. 5.

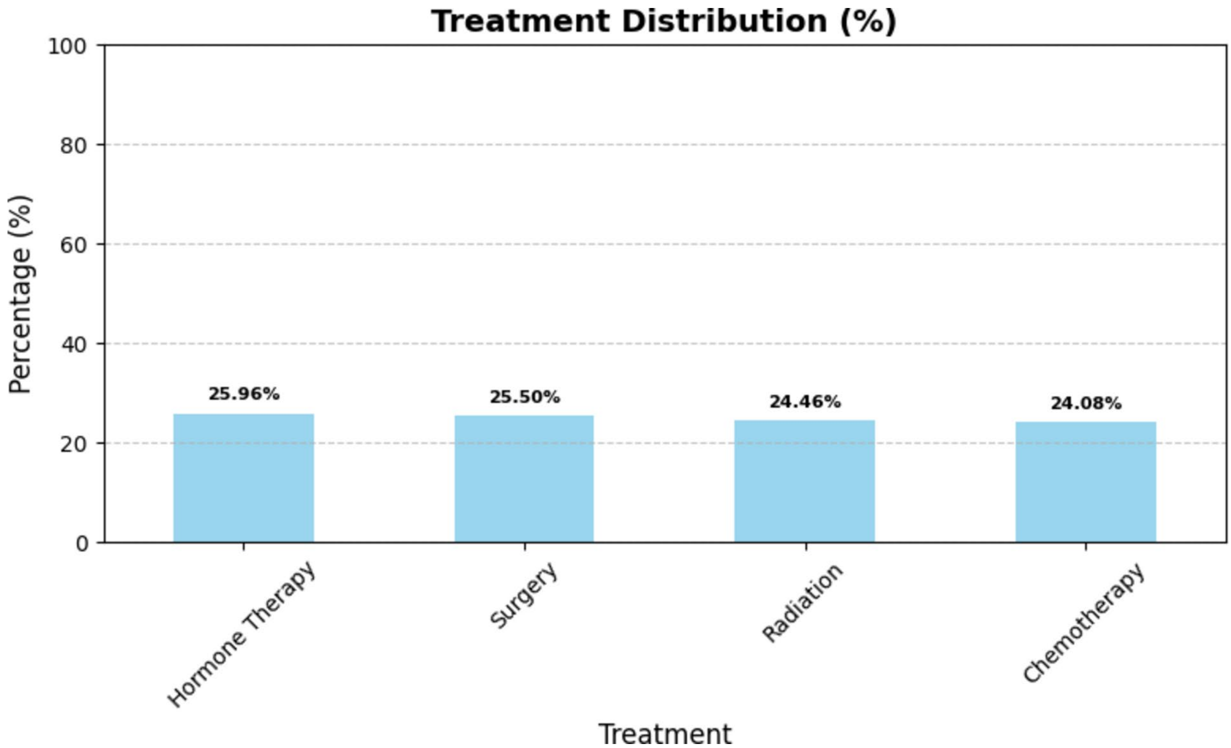

Fig. 2. Treatment Distribution (%). The percentage distribution of different treatments administered is shown in this graph. The most common is hormone therapy at 25.96%, followed closely by surgery (25.50%), radiation (24.46%), and chemotherapy (24.08%).

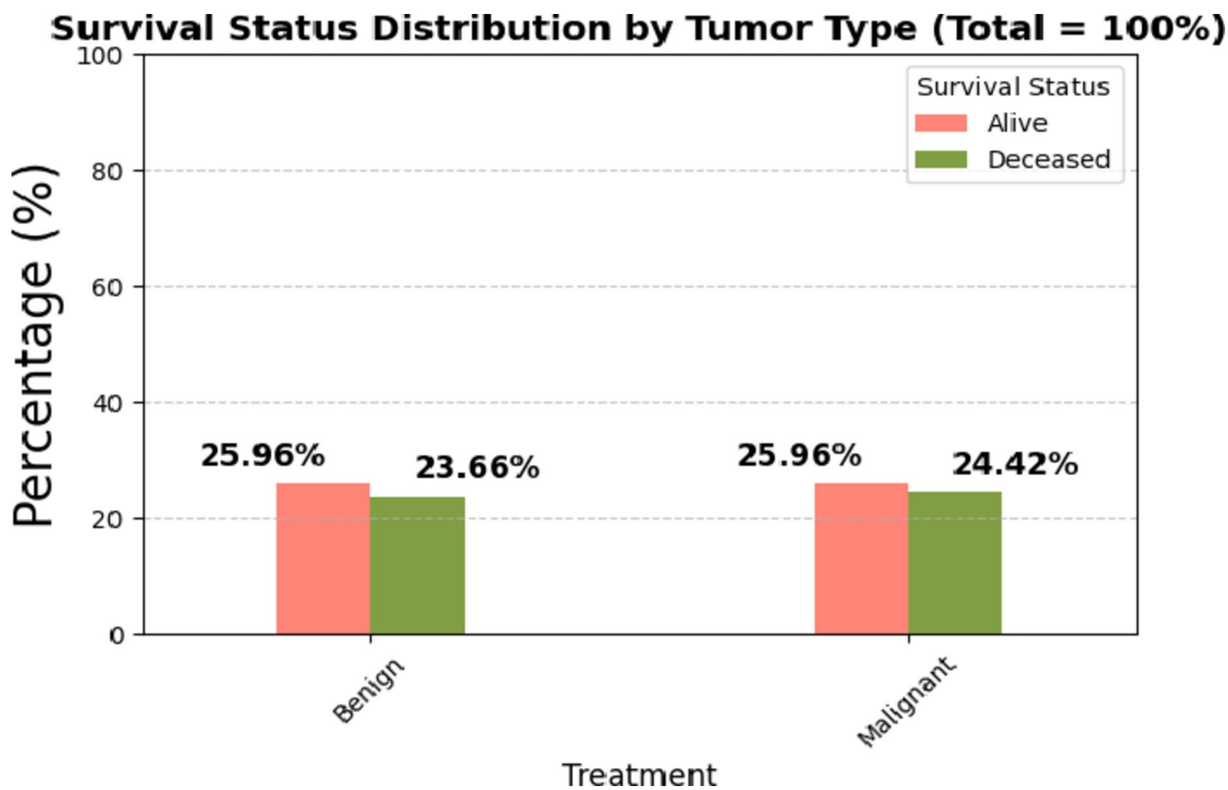

Fig. 3. Survival Status Distribution by Tumor Type. This graph divides survival status based on tumor type. For benign tumors, 25.96% are alive and 23.66% are dead. For malignant tumors, 25.96% are alive and 24.42% are dead, with both tumors showing the same survival rates.

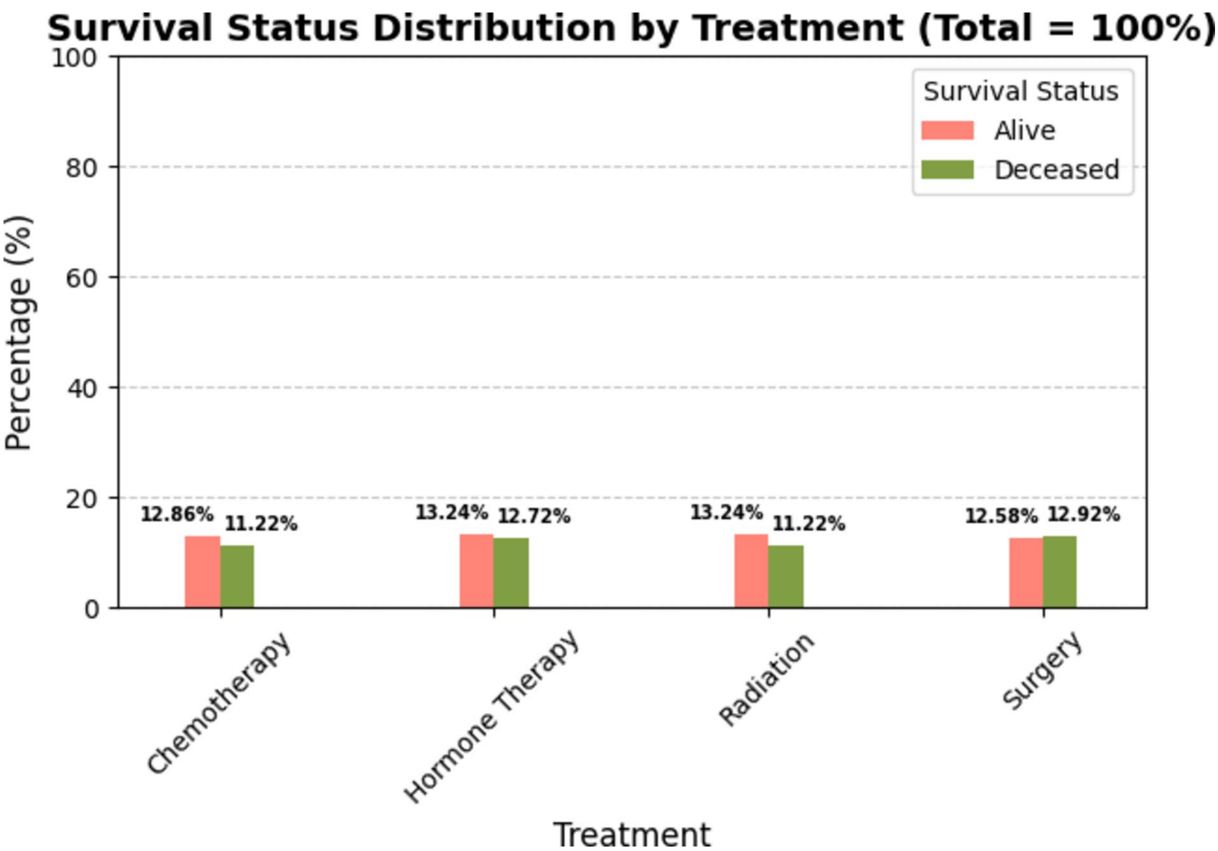

Fig. 4. Survival Status Distribution by Treatment. This graph shows survival outcomes by treatment type. Chemotherapy and hormone therapy both have slightly higher survival rates (12.86% and 13.24% alive, respectively) than radiation and surgery, which both have equal survival and deceased rates.

Feature	Test Used	*p*-value
Age	t-test	0.047
Tumor Size	t-test	0.031
Tumor Type	Chi-square	0.098
Lymph Node Status	Chi-square	0.012
Hormone Receptor Status	Chi-square	0.048
Genetic Mutation	Chi-square	0.084
Follow-Up Duration	t-test	0.023
Blood Pressure	t-test	0.041
BMI	t-test	0.037

Table 3. Statistical significance analysis of variables. This table illustrates statistical significance of various features based on hypothesis testing. The t-test for Numerical variables and chi-square test for categorical variables have been used with p-values reflecting if the features have a survival status effect ($p < 0.05$). Features such as Age, Tumor Size, Lymph Node Status, Hormone Receptor Status, Follow-Up Duration, Blood Pressure, and BMI are statistically significant but Tumor Type and Genetic Mutation are not statistically significant.

Model performance

The class distribution before SMOTE in the Alive class and in the Deceased class, making a ratio of approximately 52: 48. This ratio is generally quite balanced, but our initial performance tests showed that classifiers tended to have slightly lower recall for the minority class, Deceased. This would hint that the classifier could still be biased toward the majority class, even if only a little. SMOTE was utilized to create more representatives of the minority class during training to increase the recall without hurting the model metrics. As indicated in Fig. 6, SMOTE improves recall and F1-Score slightly, thus reducing the slight bias observed in instances where SMOTE was not applied. That would seem to suggest that SMOTE did indeed enhance the model's ability to find a pattern from the minority class without overfitting.

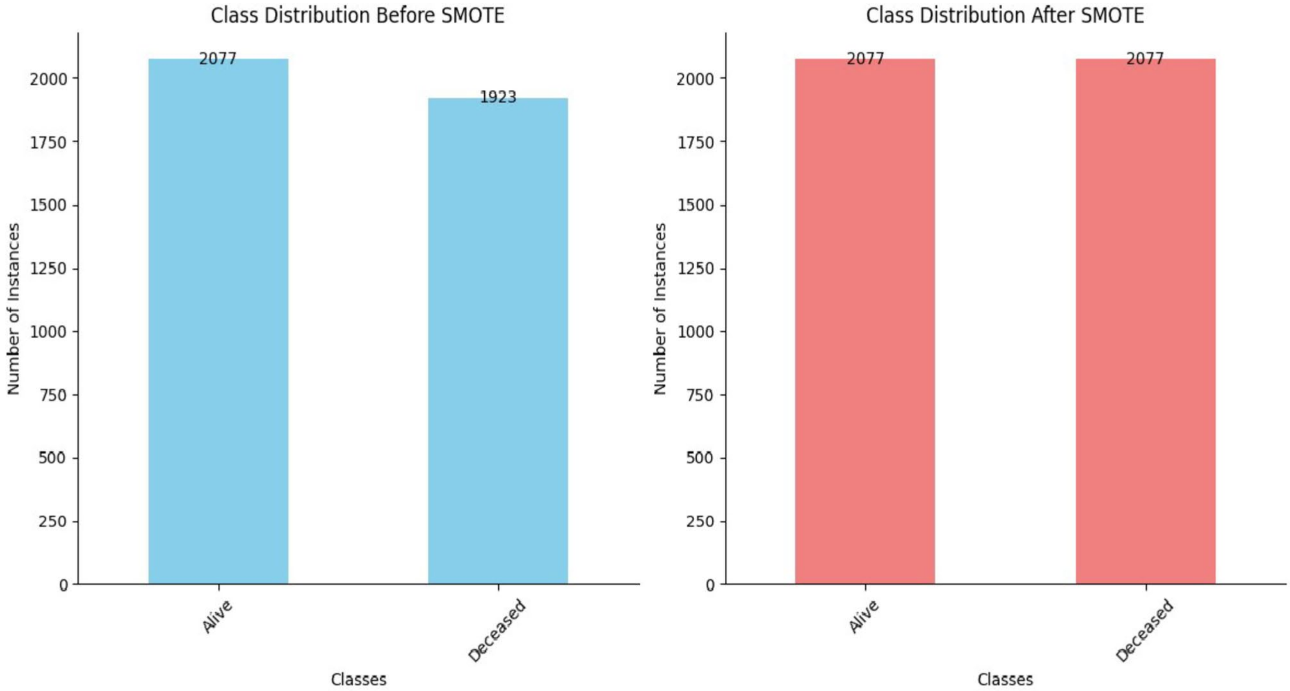

Fig. 5. This is a figure of class distribution of the target variable both prior to and after SMOTE is used. The light blue bar represents the original dataset, showing class imbalance, and the red bar represents distribution after SMOTE has generated synthetic samples in order to balance the dataset. This graphical illustration demonstrates how SMOTE accomplishes class imbalance treatment, which is crucial in improving model performance for underrepresented classes.

The heatmap of model performance metrics in Fig. 6 without SMOTE and with SMOTE showed that the Gradient Boosting was able to outperform other classifiers with Accuracy = 0.952, Precision = 0.950, Recall = 0.944, F1-Score = 0.947 and AUC_ROC Score = 0.980 without SMOTE and Accuracy = 0.972, Precision = 0.973, Recall = 0.972, F1-Score = 0.973 and AUC_ROC Score = 0.997 with SMOTE.

SHAP waterfall plot

Comparison of the four breast cancer treatment techniques using SHAP waterfall plot is shown in Fig. 7. The relative effect of chemotherapy on the f(x) or − 0.274 means that the survival predictions are not improved since the model predicts a worse survival case than the expected value. The output of the baseline model, which determines the expected value of the model E(f(x)), is -0.025, which means that the effect of chemotherapy in relation to the factors is simply negative, whereby age, tumor size, and follow-up duration are the most contributory factors.

In hormone therapy, it's -0.408 f(x), suggesting that hormone therapy causes survival prediction depression to an even greater degree than chemotherapy. Expected value E(f(x)) remains − 0.025, indicating there is additional still from age and blood pressure collar bone BMI that has the most extreme negative effect. There is better prognosis seniority among patients resorting to hormone therapy, contrary to factors likely to rescind these benefits.

Surgery has an f(x) of -0.041, and that is a bit closer to zero than the predictions of treatment. In other words, it infers that there was not much effect when determining the status of survival on that day. Such features are Follow-Up duration and tumor size—an increase in follow-up duration decreases the probability of survival prediction and is related to this prediction.

In radiation therapy, the f(x) is 0.22, showing within the range of positive values, although it also suggests a positive shift. This is more than radiation therapy has the most prediction, but still the f(x) ranges towards positive values compared to other forms. There is hardly any difference in the value of E(f(x)), which is expected to remain in the vicinity of -0.025, where Follow – up duration and hormone receptor status have the highest positive index, which means these factors are beneficial under the administration of radiation therapy.

SHAP beeswarm plot

Comparison of the four breast cancer treatment techniques using SHAP beeswarm plot is shown in Fig. 8. The chemotherapeutic SHAP beeswarm map illustrates that the model predictors cover features such as tumor size, follow-up period, and blood pressure, which appear to be the most relevant elements. The tumor size exhibits relatively skewed positive and negative SHAP values, whereas the follow-up duration is a genuine dichotomous variable.

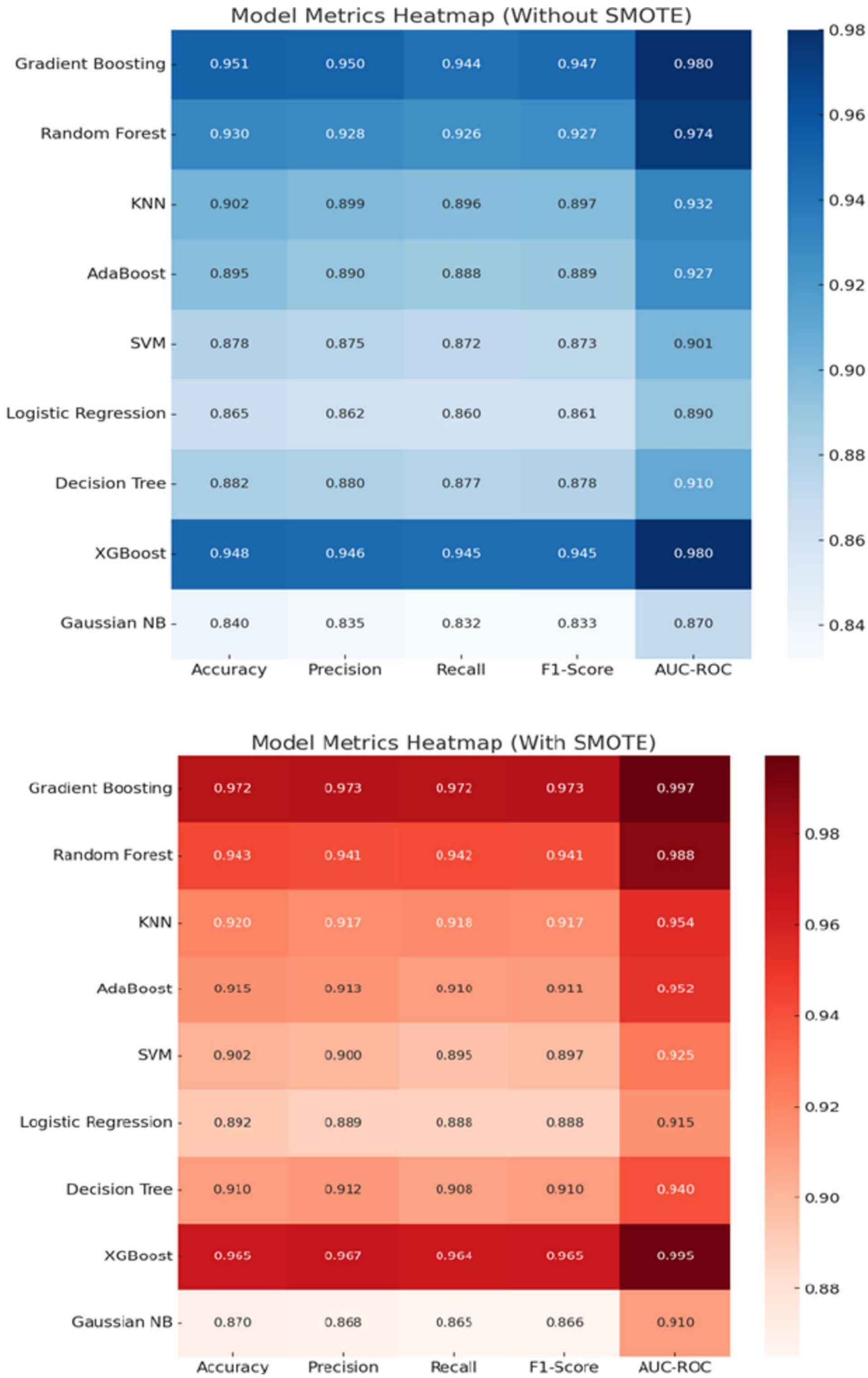

Fig. 6. Heatmap of comparison of performance of several ML algorithms on several performance metrics of evaluation both pre- and post-SMOTE application. The comparison allows direct determination of how the oversampling is affecting the classification accuracy, precision, recall, F1-score, and other metrics. The figure allows establishing whether dataset balancing enhances or worsens model generalizability.

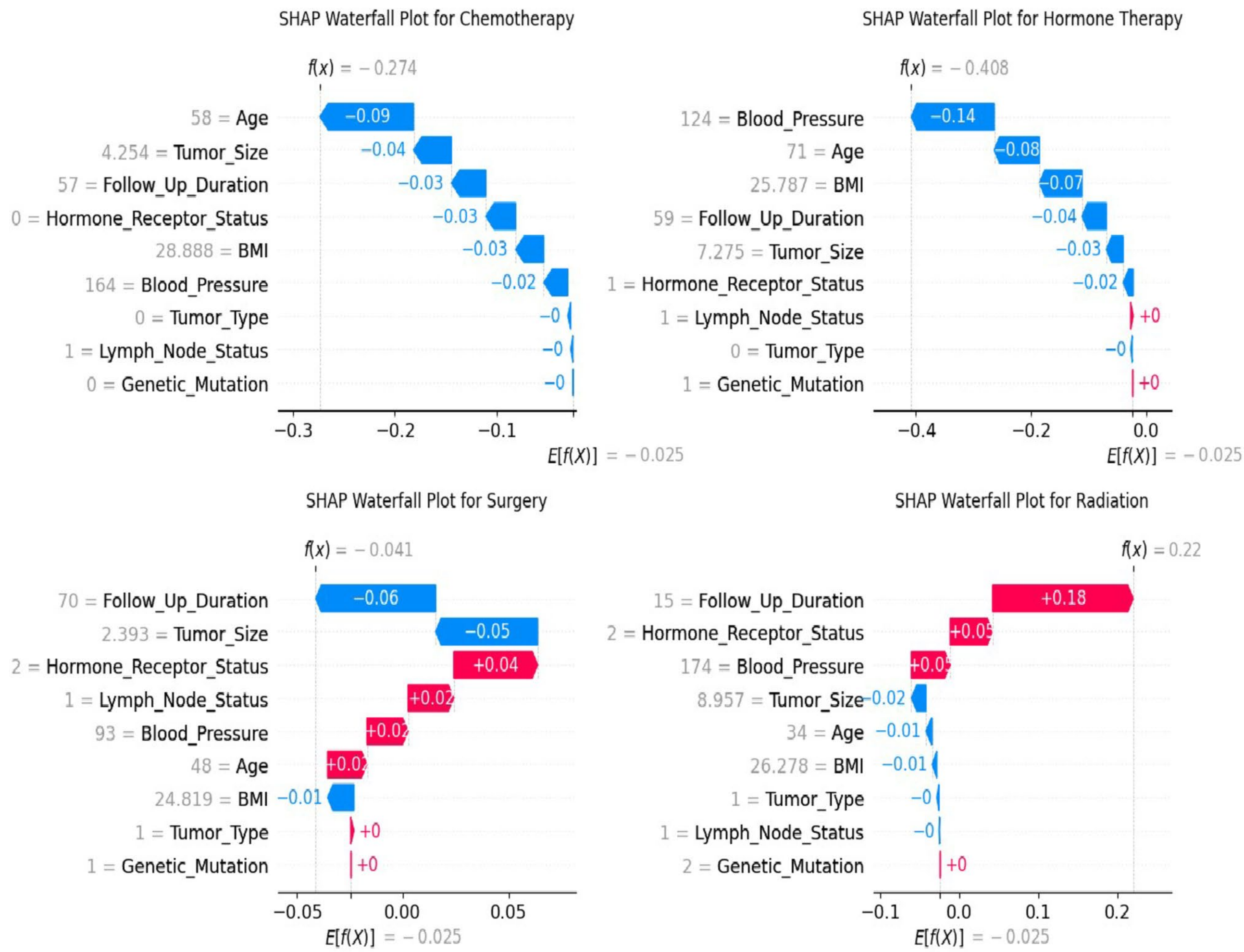

Fig. 7. A SHAP waterfall plot showing the contribution of different features to specific predictions for four treatments of breast cancer: Chemotherapy, Hormone Therapy, Surgery, and Radiation Therapy. The graph displays the influence of each feature in increasing or decreasing the model's predicted probability, facilitating a more in-depth interpretation of the influence of certain variables on decision-making for the treatment of breast cancer.

In hormone therapy beeswarm plots, tumor size, follow-up duration, and BMI are significant. Similarly, for hormone therapy, some features had a high concentration of positive SHAP scores, which indicates that these features, such as tumor size, may also adversely affect prognosis at high levels.

In the surgery beeswarm plot, the top three features defining risk are follow-up duration, tumor size, and BMI. The SHAP value distribution demonstrated the presence of factors that raised and lowered survival estimates. Considering all patients included in the model, some high follow-up duration may help while others may worsen survival.

Concerning radiation therapy, the significant features of the model are tumor size, follow-up duration, and blood pressure. On the other hand, the correspondingly larger tumor size would appear to have a strong positive SHAP value, indicating a detrimental impact on survival predictions.

Given that the patients in this cohort received only a single treatment, it reflects the unique contribution of each treatment modality to survival outcome; allowing for better understanding on their role towards overall patient survival.

Discussion

Our study reveals tracking ensemble methods like Gradient Boosting Learning model which can predict the breast cancer survival outcome with good performance. This high level of performance from Gradient Boost, evidenced by a AUC-ROC score of 0.997 — is consistent with the literature and underscores its capability for performing well on complex relationship that can arise in non-linear data relationships[12,13]. This study makes the novel contribution of integrating SHAP values into our ML model to produce a personalized analysis for treatment outcomes. We use this by attributing each feature to every prediction hence generating a more detailed interpretation on how these individual features interact with each treatment modality. The waterfall and beeswarm plots demonstrate how drastically different treatments are from one another when the effects of

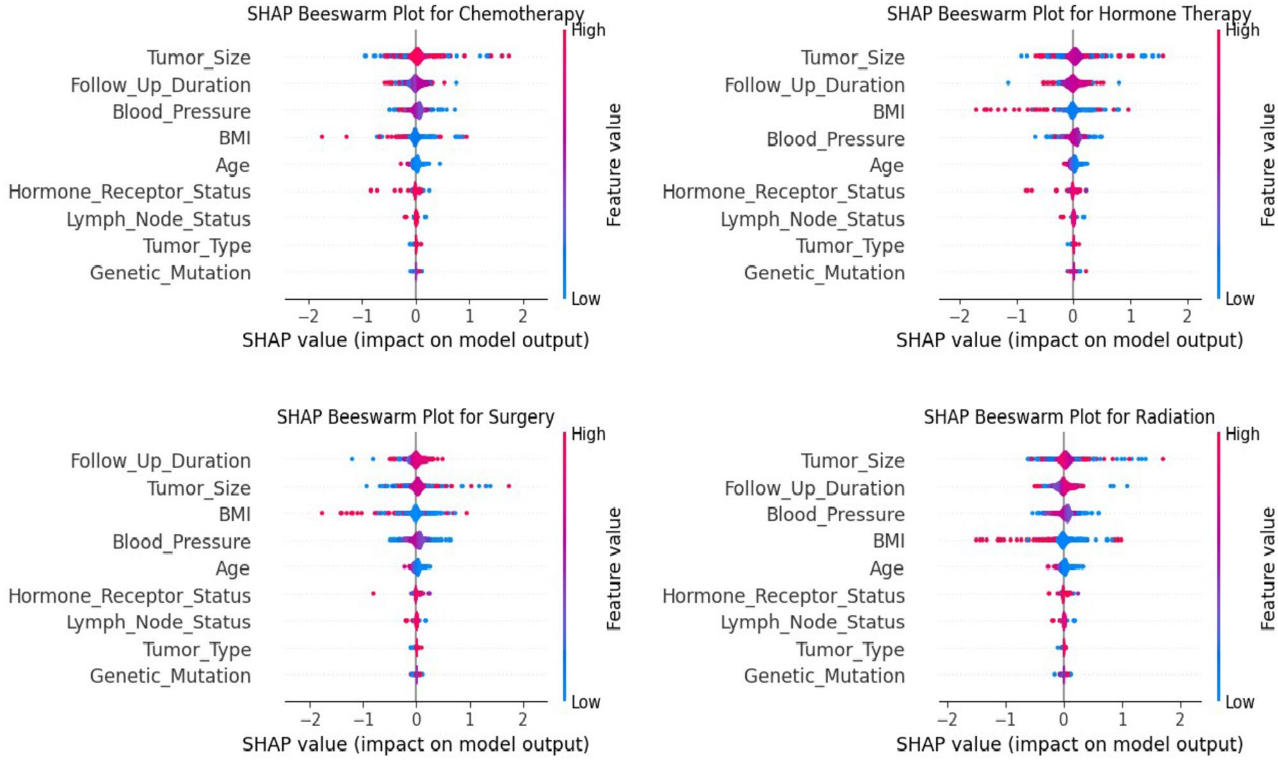

Fig. 8. SHAP beeswarm plot illustrating each feature's total contribution to model predictions across the four treatment types for breast cancer. Each point is an individual instance, and colors depict feature values. The plot reveals features that contribute significantly toward the model's output and how they cumulatively contribute to prediction variation. It helps in understanding the most significant factors determining the outcome of the treatment.

therapy are assessed on survival forecasts. The f(x) value of -0.274 in the model showed a significant negative change in chemotherapy survival outcomes when compared to the baseline's expected value of -0.025. This shows that chemotherapy has a detrimental influence on survival projections, which is in accordance with past findings demonstrating that while chemotherapy targets cancer cells, it can have catastrophic side effects that have a large impact on patient survival[14]. The combined impacts of age, tumor size, and follow-up length impact the overall picture, highlighting the complex effects of chemotherapy on patient outcomes and their compounding influence.

Hormone therapy has an even more markedly adverse effect, with a f(x) value of -0.408. This implies that the influence on survival estimates is more deleterious than chemotherapy. Age, BMI, and blood pressure all significantly contribute to this outcome, suggesting that these factors may exacerbate the negative consequences of hormone therapy[15]. This is corroborated by research, which points out that patient demographics and underlying medical issues can affect the outcome of hormone therapy, perhaps exacerbating the reported negative predictions[16].

On the other side, there are major discrepancies between how radiation therapy and surgery affect survival estimates. The minimal negative shift from surgery has a f(x) value of -0.041, which is quite neutral. This is in line with earlier studies that suggested the impact of surgery on survival outcomes would not be as great when factors like the size of the tumor and the duration of follow-up were taken into consideration[17,18]. Radiation therapy, on the other hand, demonstrates a positive shift with a f(x) = 0.22, which may be helpful for survival estimations. This conclusion is in accordance with research suggesting that radiation therapy, by focusing on cancer cells that remain after treatment, can boost survival rates[19,20]. While traditional breast cancer survival models rely primarily on tumor characteristics, genetic profiles, and therapeutic regimens, emerging evidence highlights the involvement of patient-related factors such as BMI[21–23] and Blood Pressure[24–26]. While they do not seem to be independent predictors, they influence other clinical variables (e.g., hormone receptor status, treatment type, follow-up duration) and indirectly may influence survival. Our ML model does not assume linearity but rather searches for nonlinear variable interactions, possibly uncovering novel insights into the effects of cardiovascular and metabolic variables on breast cancer outcomes.

SHAP analysis provides valuable insights into the internal decision-making process of the ML model by quantifying each feature's contribution to survival prediction. We note, however, that SHAP should not be used as an independent clinical decision-making tool. Instead, it can be used to inform understanding of risk factors, individualize treatment discussion, and enhance predictive models. For example, if SHAP indicates that hormone therapy negatively impacts survival in older patients with high BMI, this could trigger closer follow-up or alternative treatments. SHAP also helps identify potential bias in ML models to match clinical realities.

While combination therapy is the default approach for most cases, information regarding each treatment's individual effect is significant for tailored planning of treatment. The findings from our study are still able to provide hints at which treatment modalities have the most influence on survival when they are given in isolation, something that can prove useful in case monotherapy is employed due to patient-specific conditions (e.g., contraindications toward combination therapies).

However, this study is limited by potential model bias because it used a single dataset and also lacks diverse genetic and molecular data, which should be incorporated for improved performance, and lastly, the complexity of the ML algorithm without compromising clinical interpretability. Since combination therapies are standard clinical therapy, future studies would be better served using datasets of patients on multiple therapies. Further validation on more representative sets of real-world combinations of treatment is also needed.

Conclusion

The data demonstrates that one-of-a-kind cancer therapies have distinct repercussions on survival predictions; radiation therapy has a useful impact, whereas chemotherapy and hormone therapy have dreadful effects. Features together with compliance with follow-up period, tumor size, and age all play a function in the know-how of those consequences.

It is strongly advised that treatment approaches be tailored to the unique profile of every patient in light of those findings, taking into consideration the unfavorable outcomes that the model draws attention to. It is imperative that physicians frequently compare parameters that impact survival status, such as age, blood pressure, and tumor size. Further to studying techniques to lower the side repercussions of therapies like hormone therapy and chemotherapy, destiny studies ought to incorporate those discoveries into tailored treatment programs. This procedure can also boost common survival costs and enhance impacted person care.

Data availability

The data used to do this research was obtained from Kaggle dataset: https://www.kaggle.com/datasets/smmmmmmmmmmmm/breast-cancer-analysis.

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Author contributions

Authors' contributions: All authors have read and approved the final manuscript.Conception: Eyachew Misganew Tegaw, Betelhem Bizuneh Asfaw.Methodology: Eyachew Misganew Tegaw.Data collection: Eyachew Misganew Tegaw.Interpretation or analysis of data: Eyachew Misganew Tegaw, Betelhem Bizuneh Asfaw.Preparation of the manuscript: Eyachew Misganew Tegaw.Revision: Eyachew Misganew Tegaw, Betelhem Bizuneh Asfaw. Supervision: Eyachew Misganew Tegaw.

Declarations

Ethics approval and consent to participate

We hereby declare that the Kaggle dataset for this work is open-access and fully anonymized. This, in turn, means that patients' privacy will not be breached since there is no personal identifiable information in the dataset. We also affirm that the use of data does not violate any pertinent ethical standards as it is a secondary form of data collected anonymously.

Consent for publication

This manuscript is approved by all authors for publication.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to E.M.T.

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